I - Preparation and Demonstration of Slides

Note: Teacher must prepare a temporary slide using fresh specimen for demonstration during the practical hours. (If temporary slide preparation is not possible, permanent slides are allowed only during Board practical examination)

Aim: To study and identify the morphology of representative types of bacteria, fungi and Algae.

Principle: Morphology is the study of the characteristic features of the species. It could be a study of external or internal features. Morphological studies help in identification and classification of organisms.

Requirements: Buttermilk or curd, 100 ml sugar solution, crystals of yeast, bread mold, pond water, slide, cover slip to prepare temporary slides / Permanent slides of Bacteria, Yeast, Rhizopus, Chlamydomonas, Volvox, Spirogyra, Oedogonium , Compound microscope.

Exercise: 1

Bacteria *Lactobacillus*



Take sour buttermilk/curd and mount it on a slide to view lacto *lactobacillus*

Diagnostic Features

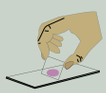
- Unicellular, Prokaryotic, rod shape, Chemo organotrophic bacteria.
- Absence of membrane bound organelles like mitochondria, nucleus, golgi bodies, plastids, etc.,
- Mesosomes are present
- Involved in lactic acid fermentation.



Figure 1: Bacteria

Exercise: 2

a. Fungi – Yeast



Add few crystals of Yeast to 100 ml sugar solution. Leave it for 2 to 3 hours. Later mount a drop of solution on a slide to view it under a microscope.

11th Botany_EM_Chapter 15.indd 342 11/01/2023 15:00:42